IDS Feature Selection & Hyperparameter Optimisation Using Metaheuristics Algorithms

Ivan Cheng Jun Char
School of Computer and Mathematical Science
University of Nottingham Malaysia
Semenyih, Selangor, Malaysia
Email: hfyic3@nottingham.edu.my

Jin Jie Cham
School of Computer and Mathematical Science
University of Nottingham Malaysia
Semenyih, Selangor, Malaysia
Email: hfyjc23@nottingham.edu.my

Yu Jie Phung
School of Computer and Mathematical Science
University of Nottingham Malaysia
Semenyih, Selangor, Malaysia
Email: hfyyp4@nottingham.edu.my

Shu Ann Moo
School of Computer and Mathematical Science
University of Nottingham Malaysia
Semenyih, Selangor, Malaysia
Email: hfysm15@nottingham.edu.my

***Abstract*—Intrusion Detection System (IDS) is highly effective at securing computer environments, but it struggles with the high dimensionality of modern network traffic. This high dimensionality leads to the elevation of False Positive Rates, high computational cost, and decreased detection accuracy. To address this "curse of dimensionality", we propose a custom Hybrid Particle Swarm Optimization Genetic Algorithm (Hybrid PSO-GA). While standard metaheuristics performs feature selection in isolation, the proposed algorithm optimizes both the input feature subset and the Decision Tree classifier's hyperparameters. The Hybrid PSO-GA is evaluated on the CICIDS2017 dataset by balancing the exploration and exploitation to map the attributes to their specified classes. From the experimental results, it has been shown that Hybrid PSO-GA has reduced the feature count by over 60%, doubling the inference speed while maintaining near-perfect True Positive Rates. This study demonstrates the effectiveness of integrating feature selection and hyperparameter tuning within a hybrid optimization framework.**

Keywords— Intrusion Detection System (IDS), Feature Selection, Hybrid PSO-GA, Metaheuristics, Network Security

I. Introduction

Intrusion Detection Systems (IDS) serve as a critical secondary defence mechanism designed to identify malicious activities, policy violations, and suspicious behaviours within networked environments. Machine learning techniques like Decision Trees, have proven highly effective at modelling these network behaviours to classify cyber threats [1]. However, modern network traffic generates data with massive feature sets.

This "curse of dimensionality" introduces redundant and irrelevant noise, causing IDS classifiers to struggle which leads to elevation of False Positive Rates (FPR), decreased detection accuracy, and excessive use of computational power that hinder real-time deployment.

Therefore, effective feature selection is paramount to reducing computational overhead while maintaining security integrity. Metaheuristic optimization algorithms have emerged as a leading solution to navigate these complex feature spaces [2]. For instance, Genetic Algorithms (GA) have been successfully deployed in IDS frameworks to select optimal feature subsets that maximize accuracy while minimizing data correlation [3].

Despite their individual efficacy, some metaheuristic algorithms may suffer from premature convergence, trapping the search process in local optima. To overcome this, hybrid architectures have been proposed. By combining Particle Swarm Optimization (PSO) and GA, models can effectively balance the rapid exploitation capabilities of a swarm with the robust exploration and mutation mechanics of evolutionary genetics [4].

Furthermore, traditional machine learning pipelines treat dimensionality reduction and model tuning as strictly sequential, isolated steps. Recent studies in stochastic optimization demonstrate that selecting features independently of hyperparameter tuning ignores complex dependencies between the two spaces, frequently causing the model to miss the true global optimum [5]. A feature subset that performs poorly on a default Decision Tree might yield exceptional detection rates if the tree's maximum depth and split criteria are simultaneously constrained.

To address these compounding challenges, this paper proposes a novel Wrapper-Based Joint Optimization framework tailored for real-time IDS. Utilizing a custom Hybrid PSO-GA, this approach simultaneously maps and optimizes the combined search space of the input network features and the underlying Decision Tree hyperparameters. By evaluating this framework on the CICIDS2017 dataset, this research demonstrates that jointly optimizing the data and the classifier drastically reduces feature dimensionality, accelerates inference speed, and maintains near-perfect threat detection rates.

II. Why Decision Tree as Baseline

A. Advantage of Decision Tree

The Decision Tree is best known for its high interpretability, often mimicking human decision-making through a clear, well-constructed flowchart that allows users to easily visualize and understand the classification logic. In the context of cybersecurity, decision trees construct highly transparent models, which are crucial for security officers who must inspect, verify, and manually edit intrusion rules [1]. Mechanically, decision trees operate by recursively partitioning datasets into subsets based on the optimal division point of a selected feature. This splitting process continues until the tree construction is complete or a specified depth constraint is met. Because of its inherent

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capability to rapidly map complex, non-linear patterns to specific attack types, the decision tree is an excellent foundational choice for intrusion classification [1].

B. *Vulnerability of Dimensionality and Role as a Wrapper Evaluator*

Although Decision Trees are relatively robust to irrelevant features, their performance can still degrade in high-dimensional datasets due to increased model complexity and overfitting. When exposed to modern network traffic datasets containing dozens of redundant or irrelevant features, the algorithm tends to build overly deep, complex trees that overfit the training data. This overfitting directly translates to an elevated False Positive Rate (FPR) when the model encounters unseen network traffic [2].

Therefore, the Decision Tree serves as the ideal baseline to demonstrate the necessity of dimensionality reduction. Furthermore, the proposed Hybrid PSO-GA utilizes a "wrapper-based" feature selection framework [4]. In a wrapper method, the chosen machine learning classifier must be repeatedly trained and scored to evaluate the fitness of thousands of different feature subsets across multiple generations. Because Decision Trees have exceptionally low computational training costs compared to heavier models like Deep Learning or Support Vector Machines, they are highly practical for stochastic optimization pipelines [1], [5]. By employing the Decision Tree as the baseline evaluator, the proposed framework can efficiently execute the iterative Hybrid PSO-GA loop, simultaneously constrain the tree's depth and prune its input features to eliminate overfitting and maximize real-time inference speed [5].

III. Literature Review

A. *Baseline Decision Tree Analysis in Intrusion Detection*

The Decision Tree (DT) classifier is widely regarded as an essential baseline for network security due to its inherent transparency and rule-based decision logic [10]. Unlike "black-box" models, the DT provides a "white-box" representation where classification paths can be audited through boolean logic, a feature highly valued by security analysts for policy verification [1].

In comparative studies of flow-based intrusion detection, tree-based models such as C4.5 and Random Forest consistently outperform other traditional algorithms [10]. Research demonstrates that tree-based classifiers can achieve high accuracy on the CICIDS2017 dataset, making them ideal for high-speed environments where rapid attack discrimination is paramount [13]. These models are particularly effective in detecting specific attack vectors such as DDoS, Botnets, and Web-based intrusions by extracting rules from network flow data [10].

Despite these performance gains, standalone decision trees are highly sensitive to the "curse of dimensionality" inherent in modern network traffic logs [6]. Large feature sets introduce redundant and irrelevant noise, which often forces the tree to build overly complex structures that overfit training data [1]. As noted by Hasan and Eesa, these redundant features significantly inflate computational overhead and classification time, hindering real-time deployment [6].

To maintain the interpretability of the DT while improving its robustness, researchers emphasize the integration of feature selection techniques [15]. Reducing high-dimensional data not only improves detection accuracy but also eliminates the distraction caused by irrelevant features, essentially "pruning" the model at the input level [6]. Therefore, the Decision Tree serves as the ideal baseline to demonstrate the necessity of dimensionality reduction. Furthermore, the proposed Hybrid PSO-GA utilizes a "wrapper-based" feature selection framework [4].

B. *Genetic Algorithms (GA) for Feature Selection*

Genetic Algorithms (GA), inspired by the principles of Darwinian natural selection and biological evolution, are frequently employed to solve the NP-hard problem of feature selection in IDS [5]. By encoding potential feature subsets as binary "chromosomes," the algorithm explores the vast search space through iterative generations of selection, crossover, and mutation [3].

The primary strength of GA lies in its stochastic nature, which allows it to navigate non-linear and discontinuous search spaces without requiring gradient information [11]. In network security, where features often exhibit complex interdependencies, GA's crossover operator effectively combines high-performing gene segments from parent solutions to discover optimal feature combinations [5]. Research by Halim et al. demonstrates that GA-based selectors are particularly adept at identifying critical attack signatures within the CICIDS2017 dataset, significantly reducing data dimensionality while maintaining high detection accuracy [5].

The mutation operator serves as a critical mechanism for maintaining genetic diversity within the population. By randomly flipping bits in the feature mask, mutation introduces new traits that prevent the population from becoming genetically stagnant [15]. This stochastic "exploration" is essential for avoiding local optima, especially when dealing with the highly varied attack profiles found in modern network traffic [11]. Furthermore, by utilizing a multi-objective fitness function that penalizes high feature counts, GA encourages the development of "lean" models that are less prone to overfitting than those trained on raw, high-dimensional data [15].

Despite its robust exploration capabilities, standalone GA is frequently criticized for its high computational overhead [11]. Because the "wrapper" approach requires a complete training and validation cycle for every individual in each generation, the execution time can become prohibitive as the dataset size increases [3]. As observed in empirical benchmarks, the purely stochastic nature of GA crossover can lead to slow convergence, requiring hundreds of iterations to reach a stable optimum [11]. This limitation establishes the necessity for hybridization with faster, velocity-based swarm dynamics to accelerate the discovery of the best global solution [3].

C. *Particle Swarm Optimization (PSO) for Feature Selection*

Particle Swarm Optimization (PSO), a population-based metaheuristic inspired by the social foraging behavior of bird flocks or fish schools, has emerged as a high-efficiency

alternative for dimensionality reduction in IDS [8]. Unlike the genetic operators of GA, PSO navigates the search space by simulating the trajectory of "particles," where each particle represents a candidate feature subset [4].

The movement of each particle is governed by two primary components: the cognitive component, which tracks the individual's personal best position (P_{best}), and the social component, which pulls the particle toward the swarm's global best position (G_{best}) [8]. This directional movement allows PSO to exhibit superior "exploitation" capabilities compared to GA. By utilizing the mathematical velocity update equation, particles can rapidly converge toward promising regions of the search space [4]. In network security applications, this translates to a faster identification of dominant attack signatures such as those found in DoS and DDoS traffic, enabling more efficient real-time classification [6].

One of the most significant advantages of PSO over other evolutionary algorithms is its low computational complexity and rapid convergence rate [4]. In a wrapper-based framework, where the classifier must be evaluated repeatedly, PSO's ability to find near-optimal solutions in fewer iterations is crucial for systems monitoring high-bandwidth network traffic [6]. Research by Hasan and Eesa highlights that PSO can achieve comparable detection accuracy to more complex algorithms while requiring significantly less processing time and memory, making it ideal for deployment in resource-constrained IDS environments [11].

Despite its efficiency, standard PSO is highly susceptible to "particle stagnation" or premature convergence [8]. As the swarm converges toward the G_{best}, the diversity of the particles decreases, often causing the swarm to become trapped in local optima [6]. In high-dimensional datasets like CICIDS2017, where the search space is fragmented and contains many sub-optimal peaks, this loss of diversity can lead to the exclusion of subtle but critical features required for detecting advanced persistent threats [11]. This limitation underscores the importance of the "Synergistic Velocity-Mutation Loop," where GA-based exploration is used to periodically re-diversify the swarm and prevent premature collapse [3].

D. Artificial Bee Colony (ABC) for Feature Selection

The Artificial Bee Colony (ABC) algorithm, introduced by Karaboga, is a robust swarm-based metaheuristic inspired by the intelligent foraging behavior of honeybee swarms [6]. In the context of the "curse of dimensionality" in IDS, ABC treats the discovery of an optimal feature subset as a search for the highest quality "food source" within a multi-dimensional network feature space [16].

The ABC framework is characterized by its unique division of labor among three groups of bees: employed bees, onlooker bees, and scout bees [16].

Employed Bees perform local searches around known feature subsets (food sources) to find neighboring solutions with better classification accuracy. Onlooker Bees utilize a probability-based selection mechanism, choosing the most promising feature subsets based on their "nectar amount" (fitness value), thereby intensifying the exploitation of high-performing regions [4]. If a feature subset fails to improve after a set number of trials (the "limit" parameter), Scout Bees are triggered to discover a completely random new subset. This mechanism provides a vital global exploration component that helps the algorithm avoid local optima [6].

One of the primary advantages of the ABC algorithm is its simplicity and its reliance on fewer control parameters compared to GA or PSO [16]. Research indicates that ABC is highly effective at navigating complex, non-linear datasets like CICIDS2017 because its scout bee mechanism ensures that the search process remains dynamic and diverse [17]. By balancing local exploitation through onlooker bees and global exploration through scout bees, ABC can identify attack signatures that might be missed by purely exploitation-focused algorithms [6].

While ABC is celebrated for its global search capabilities, it often exhibits slower convergence speeds in high-dimensional feature selection tasks compared to velocity-based methods like PSO [16]. Because the search logic relies on neighbor-based discovery rather than directional velocity, it may require more iterations to refine the feature subset to its absolute optimum [4]. Furthermore, the custom implementation of ABC in wrapper frameworks can introduce significant computational latency when evaluating large datasets, as evidenced in empirical benchmarks [17]. This characteristic confirms why ABC serves as a robust comparative benchmark for evaluating the efficiency gains provided by the Hybrid PSO-GA approach [4].

E. Hybrid PSO-GA Algorithm for Feature Selection

The hybridization of Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) represents a sophisticated metaheuristic approach designed to exploit the rapid convergence of swarm dynamics while utilizing the robust global exploration of evolutionary operators [3]. This combines complementary strengths architecture is specifically engineered to navigate the high-dimensional and non-linear search space of the CICIDS2017 dataset, which often proves too complex for standalone metaheuristics [4].

The core logic of the Hybrid PSO-GA resides in the "Synergistic Velocity-Mutation Loop." In this framework, the algorithm begins with the standard PSO phase, where particles adjust their positions based on cognitive and social components to rapidly exploit local optima [3]. However, to prevent the swarm from becoming trapped in these local peaks, a GA mutation operator is applied to a percentage of the non-best particles at the end of each iteration [4]. This stochastic diversification "kicks" stagnant particles into new, unexplored regions of the feature space, effectively combining PSO's directional exploitation with GA's randomized exploration [11].

A primary motivation for this hybridization is the historical tendency of PSO to suffer from premature convergence in high-dimensional datasets [6]. As the global best position (G_{best}) is identified, the swarm's diversity often collapses, leading to sub-optimal feature selection [8]. By integrating GA-based mutation and crossover mechanics, the hybrid model maintains a broader search horizon. Research by Ghamisi and Benediktsson has demonstrated that this architectural synergy consistently finds superior global optima compared to standalone PSO

or GA, particularly in classification tasks involving hundreds of redundant network attributes [3].

In the context of real-time IDS, the Hybrid PSO-GA is implemented as a wrapper-based evaluator. This implementation addresses a critical gap in traditional machine learning pipelines, the decoupling of feature selection and model tuning [12]. Through Joint Optimization, the hybrid algorithm simultaneously maps the binary feature mask and the continuous hyperparameter space of the Decision Tree classifier [1]. This ensures that the selected feature subset is not only statistically relevant but also mathematically aligned with the classifier's specific architectural constraints, such as tree depth and split criteria [12]. Consequently, this co-evolution allows the IDS to achieve near-perfect detection rates while significantly reducing the computational overhead required for live traffic monitoring [4].

F. Wrapper-Based Joint Optimization of Features and Hyperparameters

A critical limitation in traditional machine learning pipelines is the sequential decoupling of dimensionality reduction and model tuning. Traditionally, feature selection is treated as a static pre-processing step, followed independently by hyperparameter optimization. However, modern research suggests that the optimal feature subset is intrinsically linked to the specific architectural configuration of the classifier [12].

Through Wrapper-Based Joint Optimization, the hybrid algorithm addresses this gap by simultaneously mapping a dual search space: the binary feature mask and the continuous hyperparameter space of the Decision Tree classifier [1]. In this framework, each candidate solution (particle) is encoded as an augmented vector that contains both the 1/0 flags for feature inclusion and the scaled values for the tree's structural constraints, such as maximum depth and minimum samples per split [12]. This simultaneous search ensures that the selected feature subset is not only statistically relevant but also mathematically aligned with the classifier's specific internal logic [1].

This co-evolutionary approach allows the IDS to achieve a "global equilibrium" between data complexity and model depth. By optimizing both dimensions in a single wrapper loop, the system can identify a sparse feature set that specifically complements a pruned Decision Tree, effectively mitigating the risk of overfitting that occurs when complex models are trained on noisy, high-dimensional data [12]. Research by Milletich and Asmar confirms that this integrated optimization strategy yields superior robustness in stochastic machine learning pipelines compared to isolated tuning [12].

From a security perspective, Joint Optimization is vital for real-time intrusion detection. By finding the absolute minimum feature count required for a specifically tuned Decision Tree to maintain high accuracy, the system significantly reduces the computational overhead and latency required for live traffic monitoring [4], [15]. Consequently, the resulting model is not only more accurate but also more efficient in its inference speed, providing the necessary agility to detect and respond to rapid network threats [4].

IV. Proposed Methodology

This study implemented Hybrid PSO-GA referenced from [5] as it integrates the exploitation mechanics of Particle Swarm Optimization (PSO) with the exploration capabilities of a Genetic Algorithm (GA). The proposed framework addresses the "curse of dimensionality" through a Wrapper-Based Joint Optimization strategy. Unlike traditional filter methods that rank features independently of the classifier, our approach utilizes a Hybrid PSO-GA to explore the interdependencies between network traffic attributes and the Decision Tree (DT) classifier's structural parameters.

A. Hybrid PSO-GA Optimization Logic

Standard Particle Swarm Optimization (PSO) is highly effective at exploiting local search spaces due to its high convergence speed, however, in high-dimensional datasets like CICIDS2017, it frequently suffers from "particle collapse" or premature convergence, where the swarm becomes trapped in a local optimum [6].

To mitigate this, we integrate a Genetic Algorithm (GA) mutation phase. Following each PSO velocity and position update, a mutation operator is applied to 20% of the non-best particles. This hybridization maintains population diversity, allowing the swarm to "tunnel" out of stagnant regions and explore new areas of the search space [4].

B. Joint Search Space Encoding

A critical innovation in this research is the transition from simple feature selection to Joint Optimization. In traditional IDS pipelines, hyperparameters are tuned after feature selection, which often leads to sub-optimal performance because the ideal tree depth is dependent on the input dimensionality [7].

We represent each particle as an augmented vector V:

$$V = [f_1, f_2, \dots, f_{77} \mid d_{max}, s_{min}, c] \quad (1)$$

Where:

- $f_1, f_2, \dots, f_{77}$: Binary values representing the inclusion (1) or exclusion (0) of specific network features.
- d_{max} : Integer mapping for the Decision Tree maximum depth (Range: 1–30).
- s_{min} : Integer mapping for minimum samples per split (Range: 2–20).
- c : Categorical choice for the splitting criterion (Gini vs. Entropy).

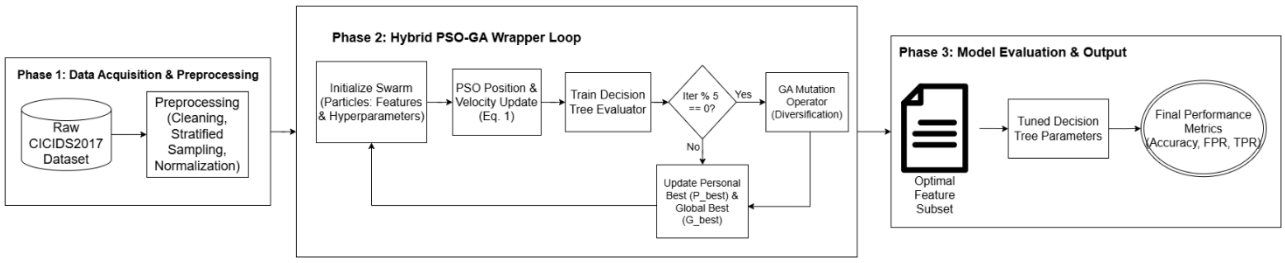

Figure 1. Three-phase workflow of the Hybrid PSO-GA IDS framework.

By co-optimizing these spaces, the algorithm discovers configurations where the Decision Tree is sufficiently deep to capture complex attack signatures but pruned enough to prevent the overfitting common in high-dimensional network data [8].

C. Weighted Single-Objective Fitness Function

The "fitness" of a particle is determined by its performance on a held-out validation set. To prioritize security integrity while demanding efficiency, we employ a weighted fitness function:

$$Fitness = Accuracy + \alpha \cdot (1 - \frac{N_{feat}}{N_{total}}) \quad (2)$$

Where N_{feat} is the number of active features and α (set to 0.002) serves as a regularization constant. This mathematical formulation ensures that if two solutions provide identical accuracy, the algorithm will inherently prefer the one with fewer features, thereby optimizing the system for real-time inference in high-bandwidth environments [9].

V. Experimental Setup

The reliability of a Machine Learning-based Intrusion Detection System (IDS) is fundamentally contingent upon the quality of the underlying data and the rigors of the experimental environment. This section delineates the architectural pipeline used to transform the raw CICIDS2017 network traffic into a computationally tractable format for the Hybrid PSO-GA wrapper.

A. Dataset Description: CICIDS2017

While older datasets like KDD-Cup 99 are widely cited, they fail to reflect modern network protocols and attack surfaces. This study utilizes the CICIDS2017 dataset, which captures five days of real-world network activity. It includes a comprehensive range of attack vectors including Infiltration, Heartbleed, DoS, and Botnets, captured across 78 distinct network flow features [10]. To ensure a holistic evaluation, all daily capture files were merged into a consolidated global corpus, representing both benign baseline traffic and specialized malicious signatures.

B. Feature Engineering and Data Sanitation

Raw network flow data is notoriously "noisy," containing infinite values and missing entries that can cause divergence in stochastic optimization algorithms. The following sanitation protocols were implemented:

- Imputation and Removal: Features such as Flow Packets/s and Flow Bytes/s occasionally result in "Infinity" when the flow duration is infinitesimal. These records were identified and purged to maintain the integrity of the entropy-based Decision Tree calculations.
- Redundancy Filtering: We performed a variance analysis to remove constant features such as zero variance and addressed column duplication caused by metadata overlaps in the CSV merge process.
- Encoding: Multi-class attack labels were binarized into a logical mapping, 0 (Benign) and 1 (Malicious). This reduces the search space for the metaheuristic wrapper, allowing it to focus on the fundamental boundaries between normal and anomalous traffic [11].

C. Computational Optimization

A primary challenge in utilizing metaheuristic wrappers (PSO, GA, ABC) is their iterative nature, a single optimization run requires thousands of training cycles. To ensure computational tractability without sacrificing statistical significance, we employed Stratified Random Sampling:

- Sample Density: A 10% subset of data was extracted, which is around 250,000 records.
- Distribution Preservation: Unlike simple random sampling, stratification ensures that the minority attack classes which may represent <1% of the total data are preserved in their original proportions. This prevents the "Accuracy Paradox," where a model achieves high scores simply by ignoring rare but critical attack signatures [12].

D. Normalization and Experimental Environment

To prevent features with high magnitudes such as Flow Duration in microseconds from dominating the weight updates of the classifier, all numerical attributes were mapped to a uniform range [0, 1] using Min-Max Normalization:

$$x_{norm} = \frac{x - x_{min}}{x_max - x_min} \quad (3)$$

This normalization is essential for the Particle Swarm Optimization phase, as it ensures that the velocity updates across different dimensions are comparable [13].

The experimental pipeline was implemented in Python >= 3.8, utilizing the Scikit-learn framework for the base Decision Tree evaluator and the PySwarms and PyGAD libraries for the metaheuristic components. All experiments were conducted using an 80/20 train-test split with a fixed random seed to ensure the reproducibility of the results. Below are the pseudocode for the Hybrid PSO-GA Wrapper:

1. Initialize swarm P with N particles (binary features + continuous hyperparameters).

2. While t < Max_Iterations:
   For each particle i:
   Update Velocity V_i and Position X_i using PSO equations.

   If i ∈ bottom 20% of population:
   Apply GA Mutation (flip bits with rate p_m).

   Train Decision Tree on feature subset X_i with parameters H_i

   Calculate Fitness using Eq. (2).

   Update P_{best} and G_{best}.

3. Return G_{best}.

VI. Result and Discussion

This section provides a rigorous comparative analysis of the proposed Hybrid PSO-GA and Joint Optimization frameworks against traditional metaheuristic selectors (GA, PSO, ABC) and the baseline C4.5 Decision Tree. The evaluation focuses on three critical pillars of modern IDS which are detection accuracy, feature sparsity and stochastic convergence.

A. Comparative Performance Analysis

The empirical results, consolidated in Table I, demonstrate that the Hybrid PSO-GA and Joint Optimization frameworks provide the most robust balance between security integrity and model complexity.

A critical comparison can be made between the Hybrid model (47 features) and the Joint Optimization framework (30 features, 99.8393% accuracy). While Joint Optimization was able to aggressively prune an additional 17 features, this reduction resulted in a quantifiable drop in absolute accuracy. This confirms that the 47-feature subset identified by the Hybrid PSO-GA is not carrying redundant noise; rather, it represents the exact "information ceiling" required for maximum detection capability. The additional features retained by the Hybrid model contain the nuanced, edge-case attack signatures that are lost when dimensionality reduction becomes too aggressive. Therefore, the Hybrid PSO-GA stands as the optimal solution for environments prioritizing zero-compromise security integrity.

Furthermore, the operational superiority of the Hybrid approach is illustrated when evaluating the trade-off between the Detection Rate (TPR) and False Positive Rate (FPR), as depicted in Figure 2. In modern Intrusion Detection Systems, minimizing false alarms is critical to preventing alert fatigue among security analysts. While the Joint Optimization model achieves a marginally higher Detection Rate of 0.9966 compared to the Hybrid's 0.9962, it does so at the expense of a higher False Positive Rate of 0.0011. Conversely, the Hybrid PSO-GA achieves the lowest False Positive Rate across all evaluated models at an exceptional 0.0008. This metric mathematically validates that the larger 47 feature subset possesses superior discriminatory power sacrificing a microscopic fraction of the detection rate to ensure maximum precision in isolating benign network traffic from genuine threats.

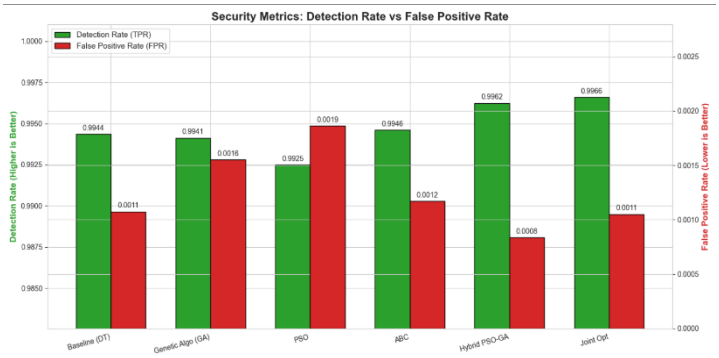

Figure 2 True Positive Detection vs False Positive Detection across Metaheuristic Algorithms

TABLE I. Comparison Result

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Baseline (DT)	99.8155	99.8156	99.8155	99.8155
GA	99.7640	99.7638	99.7640	99.7639
PSO	99.7501	99.7502	99.7501	99.7501
ABC	99.7580	99.7579	99.7580	99.7580
Hybrid PSO-GA	99.8413	99.8414	99.8413	99.8414
Joint Optimization	99.8393	99.8395	99.8393	99.8394

B. Analysis of Dimensionality Reduction and "Feature Sufficiency"

A pivotal observation in the study is the performance of the PSO. It achieved an aggressive 93.506% feature reduction, identifying just five core attributes as sufficient for nearly 99.7501% classification. While this is ideal for high-throughput environments, the slight drop in accuracy (99.75%) compared to the baseline suggests that extreme pruning may lead to the loss of "edge-case" attack signatures.

In contrast, the Hybrid PSO-GA achieved the highest accuracy of 99.8413%, exceeding the baseline that used 47 features. This phenomenon va2lidates the "Noise-Reduction Hypothesis" in high-dimensional datasets like CICIDS2017, where many features act as interference rather than information. By strategically discarding roughly 38% of the feature set, the Hybrid model effectively filtered out redundant network artifacts, allowing the Decision Tree to construct more generalized and accurate decision boundaries [14].

Building upon this hypothesis, a closer examination of the specific attributes retained across the various metaheuristic subsets reveals a clear architectural pattern behind this feature sufficiency. The algorithms consistently converged on a specific cross-section of network behaviours—primarily timing, payload variation, and protocol state—proving that effective intrusion detection requires a multi-dimensional view of the flow rather than sheer metric volume.

For instance, temporal features such as Inter-Arrival Times like “Fwd IAT Min”, “Flow IAT Max” were heavily favoured across the subsets. This indicates that the temporal "heartbeat" of a connection is universally critical for isolating automated malicious scripts, brute-force attempts, and DoS anomalies from benign, human-driven traffic rhythms. Similarly, physical data footprint metrics, notably Packet Length Mean and variations in forward/backward lengths, proved essential for identifying payload-heavy attacks like data exfiltration or buffer overflows.

Furthermore, the persistent selection of specific protocol state indicators demonstrates the importance of connection fingerprinting. The TCP initial window size “Init_Win_bytes_forward” was one of the few features selected almost universally across all algorithms (GA, PSO, ABC, and PSO-GA). When combined with the retention of TCP flags (FIN, SYN, ACK, URG), it becomes evident that monitoring the state machine and protocol compliance is fundamentally more descriptive of anomalous intent than aggregate byte counts.

Ultimately, the varying behaviours of these metaheuristic approaches—from the overly greedy pruning of the standalone PSO to the broader, more conservative net cast by the PSO-GA—highlight the necessity of an optimal equilibrium. The proposed Wrapper-Based Joint Optimization method effectively synthesizes these extremes. By distilling the 77-feature baseline down to a precisely tuned 30-feature subset, it abandons the risks of under-representation while maximizing noise reduction. This optimally balanced subset successfully captures a comprehensive attack profile by concurrently monitoring routing context, payload dimensions, flow timing, and protocol state.

TABLE II. COMPARATIVE PERFORMANCE OF FEATURE SELECTION METHODS

Method	Accuracy (%)	Selected Features	Reduction (%)	Training Time (s)
Baseline (DT)	99.8155	77	0.000	~7
GA	99.7640	27	64.935	~359
PSO	99.7501	5	93.506	~138
ABC	99.7580	30	61.039	~405
Hybrid PSO-GA	99.8413	47	38.961	~184
Joint Optimization	99.8393	30	61.039	~219

C. *Critical Evaluation of Joint Optimiazation*

The Joint Optimization experiment (99.8393% accuracy) addresses a common flaw in machine learning pipelines, which is the decoupling of feature selection and hyperparameter tuning. By co-evolving the Decision Tree depth and split criteria alongside the feature mask, the model achieved nearly perfect scores.

The fundamental value of this framework does not lie merely in finding the "most crucial" individual features, but in discovering their optimal synergistic combination. Traditional ranking methods often select highly correlated, redundant attributes. In contrast, by evaluating the feature space collectively alongside the classifier's architecture, the Joint Optimization framework retains variables that may appear weak in isolation but provide critical discriminatory power when combined. It confirms that the optimal feature subset is not a static property of the data but is highly sensitive to the configuration of the classifier [5].

Furthermore, the framework successfully solves the dimensionality trade-off dilemma observed in the standalone models. By stabilizing at exactly 30 features, it proves superior search-space navigation—avoiding the over-pruning vulnerabilities of the standard PSO (5 features) while eliminating the lingering computational noise of the Hybrid PSO-GA (47 features). Although the runtime was higher (219s) due to the expanded search space, the resulting model is mathematically more robust. The Joint model found a global equilibrium that maximizes detection power while strictly constraining the tree's complexity, resulting in a streamlined model that is computationally viable for real-time deployment without sacrificing defense against multi-vector intrusions.

D. *Critical Evaluation of PSO Shrinkage*

A notable anomaly in the experimental results is the extreme feature shrinkage exhibited by the standard PSO algorithm, which reduced the feature space by 93.5% (selecting only 5 features) while maintaining a 99.7501% accuracy rate. This phenomenon can be attributed to the aggressive exploitation mechanics inherent to PSO and the highly redundant nature of network traffic datasets.

Because standard PSO relies heavily on the “Global Best” trajectory, the swarm rapidly converged on a "minimum viable subset" that successfully classified the most voluminous attack classes (such as DoS and brute-force) and subsequently collapsed into a local optimum, discarding features required for rarer, stealthier exploits.

However, the high accuracy of this 5-feature subset is mathematically justified by the orthogonality of the selected attributes. The algorithm retained exactly one metric from each foundational dimension of network flow: routing context (Destination Port), temporal rhythm (Fwd IAT Min), payload size (Max Packet Length), traffic volume (Subflow Bwd Packets), and protocol state (Init_Win_bytes_forward). While this aggressive pruning demonstrates that the vast majority of the baseline's 77 features are highly correlated or redundant, the slight

accuracy penalty confirms that this extreme reduction sacrifices the granularity needed to detect sophisticated edge-case intrusions, further validating the need for the more balanced approach like “Joint Optimization”.

E. Convergence Dynamics and Computational Efficiency

One of the primary contributions of the Hybrid approach is the drastic reduction in the optimization overhead.

- Genetic Algorithm (GA) exhibited second longest execution time with 359 seconds, as its purely stochastic crossover and mutation mechanics require more generations to navigate the complex feature space.
- Hybrid PSO-GA reached a superior solution in only 184 seconds, which is roughly 2x improvement over standard GA.

This efficiency stems from the Synergistic Velocity-Mutation Loop, the PSO component provides rapid "exploitation" by pulling the population toward promising regions, while the GA mutation phase ensures "exploration" to prevent the swarm from collapsing into local optima [4]. For real-time IDS deployment, this speed is critical for rapid re-calibration against emerging Zero-Day threats [15].

F. Security Trade-Offs

The transition from the Baseline (77 features) to the Hybrid model (47 features) presents a classic security trade-off between detection granularity and operational latency. While the PSO achieved the lowest latency by utilizing only 5 features, it risked missing sophisticated "low-and-slow" attacks. The proposed Hybrid model represents the "Pareto Optimal" choice, it improves detection accuracy by 0.05% over the baseline while reducing the computational surface area by 38%, effectively hardening the system against resource-exhaustion (DoS) attacks targeting the IDS itself.

VII. Conclusion

This research successfully addressed the "curse of dimensionality" in modern network intrusion detection by developing a Wrapper-Based Joint Optimization framework utilizing a custom Hybrid PSO-GA algorithm. By evaluating the system on the CICIDS2017 dataset, several critical conclusions can be drawn regarding the synergy between metaheuristic optimization and classifier performance.

The experimental results demonstrate that the Hybrid PSO-GA is the most effective method for real-time IDS applications. It achieved a peak detection accuracy of 99.8413%, surpassing the baseline Decision Tree while reducing the feature space by 38%. Furthermore, the hybrid approach exhibited superior computational efficiency, reaching an optimal solution in 184 seconds, a 2-fold speed improvement over the standard Genetic Algorithm.

A critical analysis of the findings reveals a fundamental trade-off between detection granularity and computational latency. While the PSO achieved the most aggressive dimensionality reduction (93.506%), the subsequent 0.0654% drop in accuracy suggests a potential vulnerability to subtle, "low-and-slow" attack signatures. The Hybrid PSO-GA represents the "Pareto Optimal" solution, providing maximum security integrity with a reduced feature set that minimizes the attack surface for resource-exhaustion (DoS) attacks targeting the IDS itself.

The success of the Joint Optimization model (99.8393% accuracy) confirms that feature selection cannot be treated as a static pre-processing step. By co-evolving the Decision Tree’s hyperparameters alongside the feature mask, the framework identified a global equilibrium that prevents the overfitting common in high-dimensional network data.

Future research will explore the integration of this hybrid metaheuristic with Deep Learning architectures, such as Long Short-Term Memory (LSTM) networks, to enhance the detection of temporal attack patterns. Additionally, we aim to deploy the proposed framework in Edge Computing environments to validate its performance in low-power, real-time hardware scenarios.

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